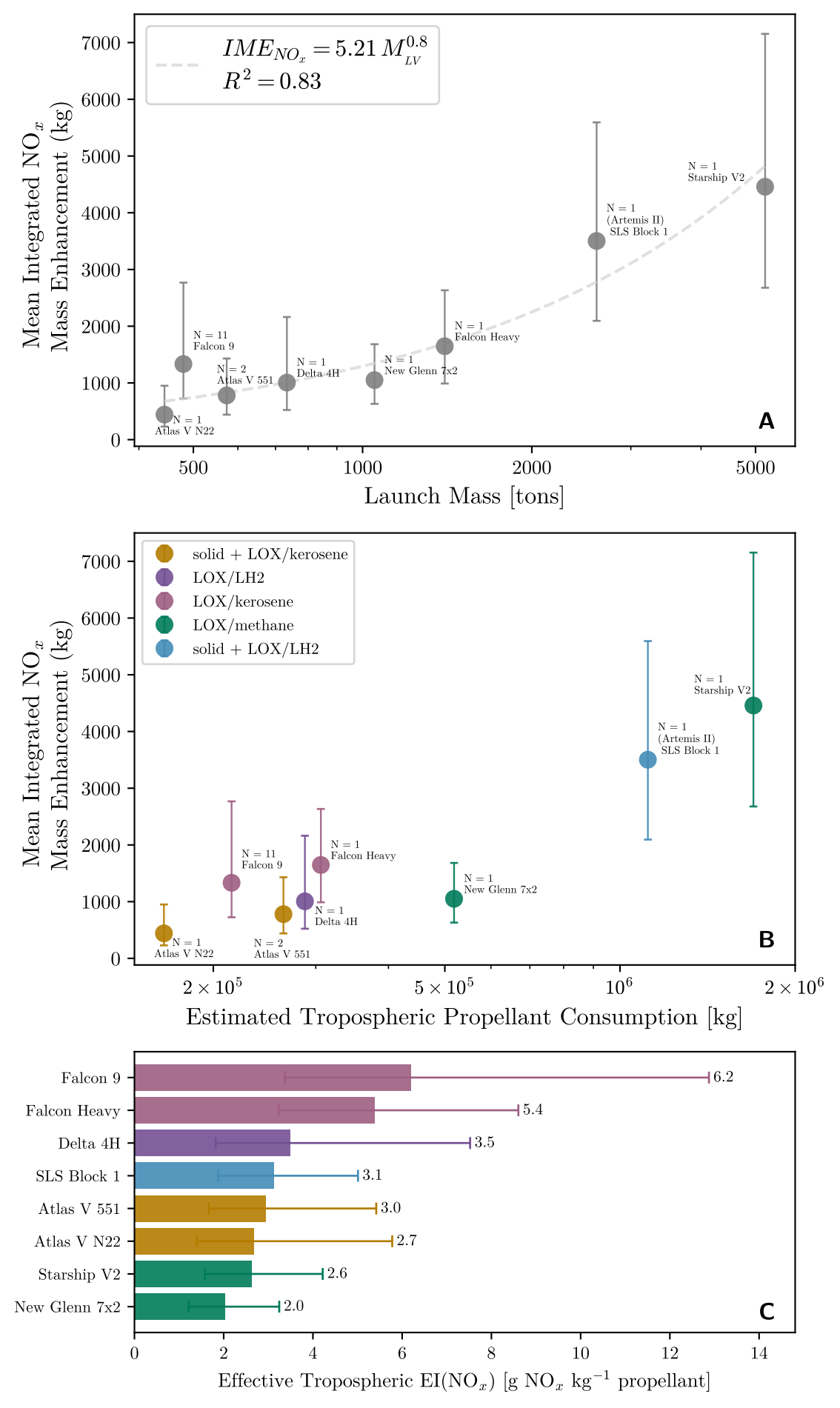
$IME_{NO_x} = 5.21\, M_{LV}^{0.8}$
$R^2 = 0.83$
Mean Integrated NO_x Mass Enhancement (kg)
Launch Mass [tons]
N = 1 Atlas V N22
N = 2 Atlas V 551
N = 11 Falcon 9
N = 1 Delta 4H
N = 1 New Glenn 7x2
N = 1 Falcon Heavy
N = 1 (Artemis II) SLS Block 1
N = 1 Starship V2
A
solid + LOX/kerosene
LOX/LH2
LOX/kerosene
LOX/methane
solid + LOX/LH2
Estimated Tropospheric Propellant Consumption [kg]
B
Falcon 9
Falcon Heavy
Delta 4H
SLS Block 1
Atlas V 551
Atlas V N22
Starship V2
New Glenn 7x2
6.2
5.4
3.5
3.1
3.0
2.7
2.6
2.0
Effective Tropospheric EI(NO_x) [g NO_x kg^{-1} propellant]
C